Curve25519 index calculus, wave 5: thirteen new leads, one law found

in the data (2026-11-12) **Class.** Lead proposals and desk checks. Nothing was run on a curve except reading committed cell files; no ecbench session, no row, no leaderboard or atlas change. No pcc claim was registered, so every statement is an unverified assertion [pcc:unverified].

Owner's request (2026-11-12). "Devise a bunch of new leads for Curve25519 index calculus, find some new structure or something to exploit."

How. Three lanes worked in parallel from the same briefing. Their full records are in `lanes/`:

- `real_parameters.md`: structure of the real curve's p, A, λ and ℓ (idea-generator, 5 records).
- `pipeline.md`: whole-pipeline cost of the current baseline (idea-generator, 4 records).
- `synthesis.md`: connections to other fields (idea-synthesist, 4 records and 8 rejections).

The main session checked the lanes' arithmetic and code citations, and tested one law against the committed data (`checks/`). Lanes had no shell, so every figure a lane derived by hand stays unchecked unless this README says it was checked.

1. Answer

- **A law in the existing data.** The Gröbner solving degree of a decomposition system on a subgroup base V = μ_d tracks the *largest prime factor* e_max of d, not |V| = d. It equals d + 1 at every prime d (20 of 20 recorded cells). It is e_max + 2 to 3 at composite d with m = 2, and e_max + 3 to 5 with m = 3. The synthesis lane preregistered eight k = 22, m = 3 values that it states it had not read; all eight match the recorded cells (`checks/sd_law_check.out.txt`). The lane's statement that it had not read them cannot be verified [pcc:unverified].
- **What the law says about the real curve.** p − 1 = 4 · 3 · 65147 · C, with C a 236-bit probable prime (checked here). Every μ_d base of Curve25519 with d ∤ 12 therefore has e_max ≥ 65,147, and so solving degree at least 65,148 on that route. The useful |V| window for the PKM attack, about 2^64 to 2^85, holds no divisor of p − 1 at all. This is a security statement for the subgroup-base route, not an attack [pcc:unverified].
- **Measured-improvement candidates for the baseline.** Four pipeline levers on the walk-collision IC arms could each give a measured gain at k = 20 and 24. The largest prediction is about 0.83× cost at k = 20 for recycling set-up points. None changes an exponent. After all four, the C = 2 track would sit about 1.7 to 1.8× above rho-neg at k ≥ 28, which is the distance remaining [pcc:unverified]. The owner ruled on 2026-11-04 that these arms count as index calculus (`research/curve25519_autolab_20261104/README.md`).
- **Algebraic-track candidates.** Two levers target the cost of one decomposition system: a flat membership presentation and a coset rule from the descent character. They improve a component, not the baseline, because at m ≤ 3 plain enumeration on the same base is still cheaper than F4 (pipeline lane, §0).

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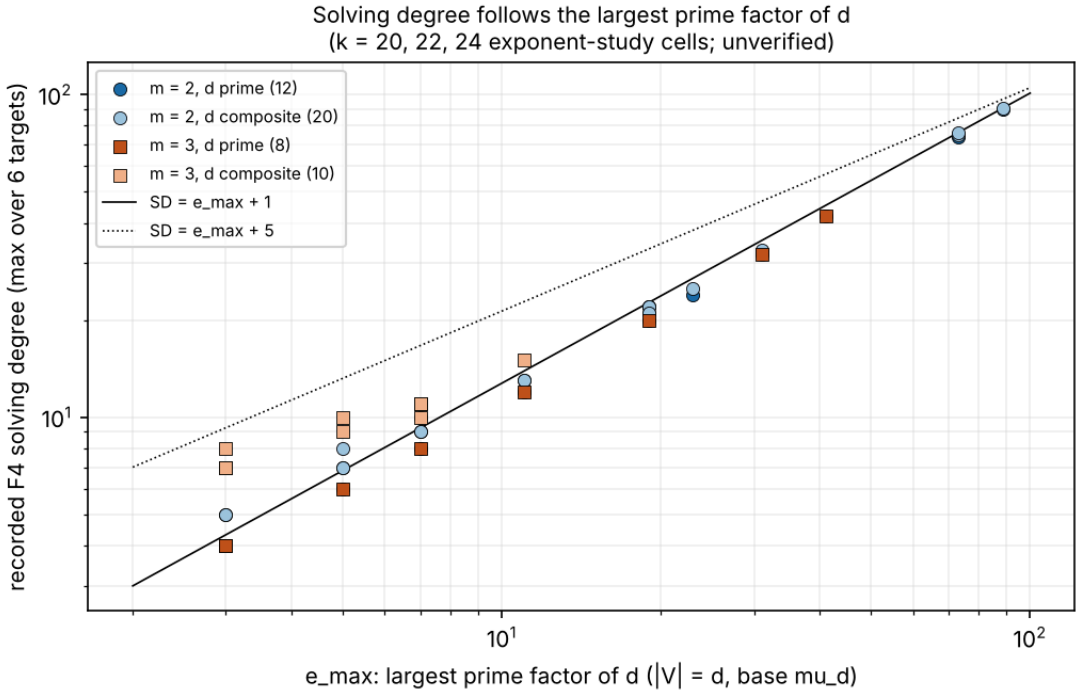


Figure 1: Recorded F4 solving degree against the largest prime factor of d on all 50 complete exponent-study cells (k = 20, 22, 24; m = 2, 3). Unverified.

2. Ranked leads

The rank orders leads by expected measured improvement per unit cost, then by how decisive they are [pcc:unverified]. "Improvement" leads could produce a verified gain over the lab's IC baseline. "Security" leads explain why the real curve resists. "Prior" is the lane's stated probability that the prediction holds.

#	lead	class	prediction	cheapest test	prior
1	W5-PL-3 final [d]G = Q check from the stored [2^i]G ladder	improvement, pipeline	−2 % at k = 20, −0.7 % at k = 24; walks unchanged, so paired runs differ only by the check	hours of Rust; minutes of paired runs	0.85
2	W5-PL-1 store every set-up point in the collision table	improvement, pipeline	0.83× at k = 20, 0.95× at k = 24, none at k ≥ 28; never worse at w = 0	about 1 day of Rust; minutes of runs; needs a rho reference-parity control	0.75
3	W5-SYN-3 solving degree = largest prime factor law	security + instrument	prime d: SD = d + 1; composite: e_max + [2, 3] (m = 2), + [3, 5] (m = 3)	done in-sample here (50 cells, 8 of 8 preregistered); next: k = 24, m = 3 and a Lean or CAS argument for the prime case	0.8
4	W5-RP-1 flat y^d = 1 per summand instead of the power-map chain	improvement, algebraic component	flat cost lies on the prime-d curve; chain / flat 70 to 260× at d = 27, 28 (k = 20, m = 3); d = 36 completes under 3.5 GB	under 1 h; chain cells already on disk	0.65
5	W5-SYN-1 descent-character coset rule for μ_d bases	improvement, algebraic component	gidx 0 needs 2× the solves per relation at m = 3; control m = 2 at 1×	about 10 min census, under 1 CPU-h	0.6
6	W5-PL-2 walk that adds a random recent stored point	improvement, pipeline	−3 to −4 % at k ≥ 28 (the only lever there), about −22 % at k = 20, overlapping PL-1	stage-0 simulation in `src/bin/nc2_stage0.rs`, no group operations	0.5
7	W5-RP-4 split sparse/dense F4 linear algebra	instrument (memory)	peak memory ≤ 1/4 of dense F4; k = 20, m = 3, d = 36 completes within 12 GB, so the frozen exponent estimator can apply	about 1 day of Rust, up to 8 h of runs	0.4
8	W5-PL-4 jump count at the law's optimum	improvement, configuration	8 / 16 / 30 / 57 jumps at k = 20 / 24 / 28 / 32; 0.983× at k = 32	configuration only	0.4
9	W5-SYN-2 split of the yield heuristic at the Weil threshold	security + corpus instrument	mean yield is a theorem: 2× a random base on the right coset, 0 on the other	shares the census tool of #5	0.6
10	W5-RP-2 binomial versus dense versus random base of equal size	security	dense / binomial cost 1 to 8; random / dense 0.5 to 2	under 2 h	0.55
11	W5-RP-3 torus and cyclotomic-torus bases for Curve25519	security	no 2^32-smooth divisor ≥ 2^64 of Φ_k(p) for k ∈ {1, 2, 3, 4, 6}; torus base costs as a random base	desk census plus minutes	0.6
12	W5-SYN-4 sumset base x(A ⊕ B) on an auxiliary curve	improvement, low prior	wins only for 60 ≲ d ≲ 1,500 at m = 3; needs no smooth p ± 1	about 1 day, ≤ 6 CPU-h	0.25
13	W5-RP-5 CRT tensor presentation μ_{d1 d2} = μ_{d1} × μ_{d2}	improvement, low prior	beats the chain on ≥ 2 of 3 cells; never beats flat	under 1 h	0.15

3. How the leads connect

- **#3 and #4 test the same tension from two sides.** The chain presentation lowers the per-variable degree to e_max but adds one variable per prime factor of d. In the exponent study, cost rose about 8.5× per prime factor (validator joint fit). The flat presentation keeps one variable but has degree d, so by #3 it should have solving degree d + 1 even for composite d. #4 measures which side wins. A flat win would also remove the only reason the μ_d route needs smooth p − 1. A chain win would confirm the SD law as the cost driver.
- **#5 overlaps an earlier lab result.** The lab's T_0-symmetry study (`research/curve25519_t0_symmetry_20261102/`) found that x ↦ 1/x, the action of the 2-torsion point, doubles coverage for pair tables. #5 is the same symmetry seen from the algebraic side: with g = 1 the base μ_d is closed under x ↦ 1/x, so each relation is found 2^(m−1) times. #5 should cite that study, and its census must show the duplicates are not already removed by the engine.
- **#1, #2, #6 and #8 all act on the walk-collision arms.** #2 and #6 overlap at k ≤ 24 and their gains do not add. #8 is moot if #6 lands. The pipeline lane's desk finding behind #6 and #8 explains the measured 3 % walk excess as the repeat rate of a

stored walk with 16 fixed jumps: (1 − 1/16)^(−1/2) = 1.0328 (checked here). That arithmetic is checked; the fit to the walk data is the lane's and is not checked.
- **#7 is the instrument the exponent question is waiting for.** The m = 3 verdict stayed "undetermined" because d = 36 ran out of memory at 12 GB (`research/curve25519_exponent_study_ext_20261110/`). #7 is the only lead here that could let the frozen estimator apply.

4. Structure looked at and rejected

The real-parameter lane rejected these as levers, with reasons in its file: the small-height λ = 121665 / 121666 of the 2-isogenous Legendre curve, the small A, the near-power-of-two ℓ = 2^252 + c, the carry structure of 2^255 − 19, NTT evaluation over a subgroup, μ_4, E′′s full 2-torsion, cofactor 8, and the small 2-adic valuations of p ± 1. The synthesis lane rejected Guruswami–Sudan list decoding, sum-product bounds, BKK mixed volumes, Kani–Robert torsion attacks, Blahut linear complexity and lift-fraction coset selection. The pipeline lane closed these directions for the current baseline: linear algebra (about 10 row operations per run), large primes and cycle-finding (each large prime is another unknown), multi-target reuse (excluded by the owner), factor-base size (autolab grids are flat), and Montgomery batching (no change in counted GAE) [pcc:unverified].

5. Checks done in this session

check	result	file
p − 1 and p + 1 factorisation by trial division plus a Fermat test	p − 1 = 4 · 3 · 65147 · C, C 236-bit probable prime; p + 1 = 2 · 5^2 · 79 · M, M composite (244 bits). The brief's "C ≈ 221 bits" was wrong, as the lane said	session log
2-isogenous Legendre curve on the k = 20 miniature	E′ = E/⟨(0,0)⟩ has #E′ = #E and full 2-torsion; λ = 231/232 mod p; the Legendre model is the quadratic twist of E′ (point count 2p + 2 − #E′)	session log
SD law, all complete sym cells at k = 20, 22, 24	prime d: SD = d + 1 in 20 of 20; preregistered k = 22, m = 3: 8 of 8	`checks/sd_law_check.py`, `.out.txt`
Pipeline lane code citations	`plugins.rs` walk parameters, `ic_boundary.rs` jump draw with γ uniform in [1, r), and the `g.mul` recovery check all exist as cited	session log
Owner ruling on walk arms	C = 2 seed arms and walk arms with known-log G jumps count as IC (2026-11-04)	`research/curve25519_autolab_20261104/README.md` line 99

6. Not done and caveats

- research-kb, ecdb and the lead atlas bodies were not reachable from this runtime. Novelty is checked only against the 176 atlas lead ids and the repository files the lanes read. The nearest atlas ids are named in each lane record. A knowledge-base pass is owed before any lead becomes a task card.
- Every lane number derived by hand is unchecked unless §5 lists it. Most literature was read from abstracts. The pipeline lane owes a literature check on "self-sum walk" and "recycled set-up points" (Teske; Bernstein–Lange, ePrint 2012/294).
- No experiment was run, so there is no benchmark record, Conductor task or resume checkpoint.
- The lanes ran on a different model tier from the agents' defaults, because the default tier's weekly limit was reached. That changes no rule.

7. Next steps

The owner decides which to run. In the order of §2, the cheapest pair with a likely measured gain is #1 then #2. Both need a frozen protocol, a paired single-target ecbench session, a validator and a matched rho reference under the §5a band. The cheapest decisive desk item is #3 at k = 24, m = 3, using cells already on disk once they exist.

Files

file	content
`lanes/real_parameters.md`	5 records on the real curve's parameters (W5-RP-1 to 5)
`lanes/pipeline.md`	4 records on the walk-collision pipeline (W5-PL-1 to 4)
`lanes/synthesis.md`	4 cross-field records (W5-SYN-1 to 4) and the rejections
`checks/sd_law_check.py`, `checks/sd_law_check.out.txt`	the SD-law check on committed cells
`checks/sd_law_figure.py`, `sd_law.svg`, `sd_law.png`	the figure
`REPORT.typ`, `REPORT.pdf`	this README as a report